# Functional Requirements

**Status:** Requirements Baseline **Version:** 1.0 **Priority:** MoSCoW

## 1. Primary System Actors

**Security Operator:** Investigates incidents & manages alerts. | **Cloud Admin:** Manages AWS test resources. | **IoT Admin:** Controls IoT devices & policies. | **Automated Response Engine:** Executes containment rules.

## 2. Requirements Register (FR-01 to FR-10)

| ID | Requirement Description | Priority | Acceptance Evidence |
|---|---|---|---|
| **FR-01** | Centralized dashboard for findings, devices, and incidents. | **Must** | UI renders seeded & live test data. |
| **FR-02** | Represent IoT devices with unique ID & recorded state. | **Must** | Device registry displays unique IDs. |
| **FR-03** | Ingest cloud events & MQTT telemetry streams. | **Must** | Test event passes ingestion pipeline. |
| **FR-04** | Identify defined suspicious behaviors via rules. | **Must** | Positive test cases flag findings. |
| **FR-05** | Retain time, source, entity, and event details. | **Must** | Required metadata fields saved in DB. |
| **FR-06** | Calculate explainable risk score based on indicators. | **Must** | Unit tests verify calculation logic. |
| **FR-07** | Assign severity label based on score thresholds. | **Must** | Scores map to Low/Med/High/Critical. |
| **FR-08** | Correlate multi-source events via matching logic. | **Must** | Related events aggregate into 1 incident. |
| **FR-09** | Create/update incident records for findings. | **Must** | Unique Incident ID generated. |
| **FR-10** | Display incident timeline & evidence references. | **Must** | Operator opens incident detail view. |

## 2. Requirements Register (FR-11 to FR-20)

| ID | Requirement Description | Priority | Acceptance Evidence |
|---|---|---|---|
| **FR-11** | Trigger automated response workflow on threshold match. | **Must** | Qualifying incident triggers response. |
| **FR-12** | Restrict automated containment to allowlisted targets. | **Must** | Out-of-scope targets ignored. |
| **FR-13** | Log response attempts, outcomes, and error logs. | **Must** | Audit trail updated with execution state. |
| **FR-14** | Provide documented device/resource recovery process. | **Must** | Entity restored via documented steps. |
| **FR-15** | Preserve incident records for investigation. | **Must** | Closed incidents remain queryable. |
| **FR-16** | Filter & review incidents by severity and status. | **Should** | Dashboard supports status filtering. |
| **FR-17** | Display chronological incident timeline view. | **Should** | Timeline matches event timestamps. |
| **FR-18** | Audit configuration changes affecting rules. | **Should** | Config edit events logged. |
| **FR-19** | Provide synthetic event demonstration mode. | **Should** | Demo scenario runs without external target. |
| **FR-20** | Export incident details report. | **Could** | Incident summary export option. |

## 3. Core System Use Cases

### UC-01: Security Overview Review

Operator opens dashboard, views threat indicators, and inspects high-risk incident detail views.

### UC-02: IoT Anomaly Detection

Device emits abnormal MQTT traffic; rule triggers finding and updates device risk score.

### UC-03: Event Correlation

Engine correlates device failure with cloud trail event and aggregates into single incident.

### UC-04: Automated Containment

High-risk score triggers auto-containment, applying restrictive IoT policy to isolated target.